University of Warsaw
Faculty of Economic Sciences

Markiian Strohyi
Album N^{o}: 449871

Forecasting Volatility and Risk with a Dynamically Weighted GARCH-LSTM Ensemble Model

Licencjat degree thesis
Field of the study: Finance, International Investment, and Accounting

The thesis written under the supervision of
dr. Paweł Sakowski and dr. Jakub Michańków
Faculty of Economic Sciences, UW

Warsaw, 2025

Declaration of the supervisor

I declare that the following thesis was written under my supervision and I state that it meets all criteria to be submitted for the procedure of academic degree award.

*I declare that my participation in the scientific article, which is a part of this thesis is 20%, while the supplement to the thesis was written independently by the graduate(s).

** cross out if not applicable*

Date | Signature of the Supervisor

PODPIS ZAUFANY
JAKUB KONRAD
MICHAŃKÓW
29.05.2025 10:45:37 [GMT+2]
Dokument podpisany elektronicznie
podpisem zaufanym

Podpisany elektronicznie przez
Paweł Marcin Sakowski
29.05.2025
10:41:21 +02'00'

Declaration of the author of the thesis

Aware of the legal responsibility, I declare that I am the sole author of the following thesis and that the thesis is free from any content that constitutes copyright infringement or has been acquired contrary to applicable laws and regulations.

I also declare that the thesis has never been a subject of degree-awarding procedures in any higher education institution.

Moreover I declare that the attached version of the thesis is identical with the enclosed electronic version.

* I declare that my**** participation in the scientific article, which is part of this thesis is 80% (not less than 60%), while the supplement to the thesis was written by me***.

** cross out if not applicable*

Date | Signature of the Author

PODPIS ZAUFANY
MARKIIAN
STROHYI
29.05.2025 10:27:17 [GMT+2]
Dokument podpisany elektronicznie
podpisem zaufanym

** *each of the co-authors of the thesis submit the statement separately.*
**** *in the case of co-author of thesis, substantive and percentage contribution should be declared.*

Abstract

This thesis investigates the out-of-sample performance of three volatility forecasting models namely Generalized Autoregressive Conditional Heteroskedasticity (GARCH), Long Short-Term Memory (LSTM) networks, and a dynamically weighted GARCH-LSTM ensemble, in forecasting volatility and estimating market risk using daily S&P 500 return data from 2004 to 2024. Log returns were computed from Yahoo Finance closing prices, and squared returns were used as a proxy for volatility. The GARCH model was estimated using a rolling window approach with a 252-day estimation window and 20-day refitting frequency. The LSTM model was trained using a walk-forward validation strategy with a 1008-day training window and 504-day test window. The ensemble model assigned dynamic weights to GARCH and LSTM forecasts based on the inverse of recent Mean Absolute Error (MAE) values, allowing adaptive re-weighting over time.

Forecast accuracy was assessed using error metrics including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Heteroskedastic MSE (HMSE). Risk estimates were evaluated through Value-at-Risk (VaR) and Expected Shortfall (ES), with reliability assessed via backtesting using the Kupiec Proportion of Failures (POF) test, the Christoffersen Conditional Coverage test, and the McNeil-Frey test for ES.

Empirical results show that although the LSTM model achieved lower forecast errors compared to GARCH, it produced the highest number of VaR violations and failed key risk backtests. The ensemble model consistently outperformed both GARCH and LSTM in terms of forecast accuracy and showed improved performance in some risk diagnostics, but it did not fully meet all backtesting criteria. The GARCH(1,1) model, while yielding higher forecast errors, demonstrated the most reliable performance in tail risk estimation and came closest to satisfying the statistical requirements for VaR coverage.

Overall, the findings suggest that while machine learning models like LSTM enhance forecast precision, this does not necessarily translate into superior risk estimation. The ensemble approach offers a promising middle ground, combining the strengths of statistical and data-driven models. This research contributes to the literature on hybrid financial forecasting and highlights the trade-offs between accuracy and statistical reliability in risk-sensitive applications such as VaR and ES estimation.

Key words

Time series modeling, S&P 500 volatility, dynamic ensemble forecasting, GARCH-LSTM integration, conditional risk estimation, VaR and ES backtesting, machine learning in finance

Field of the thesis (codes according to the Erasmus program)

Economics (14300)

The title of the thesis in Polish

Prognozowanie zmienności i ryzyka za pomocą modelu GARCH-LSTM z dynamicznymi wagami.

Contents

1 Introduction **5**

2 Literature Review **6**

3 Dataset description **10**

4 Methodology and Empirical Framework **13**
4.1 GARCH 14
4.1.1 Model Choice and Configuration 14
4.1.2 GARCH Realized Volatility Prediction Approach 16
4.2 LSTM 16
4.2.1 LSTM Architecture 17
4.2.2 Hyperparameter Tuning and Model Setup 18
4.2.3 LSTM Forecasting Approach 19
4.3 GARCH-LSTM Ensemble Model 20
4.3.1 GARCH-LSTM Forecasting Approach 20
4.4 VaR and ES 22
4.5 Model Evaluation Metrics 23

5 Results and Error Metrics **25**
5.1 Evaluation of GARCH Model Performance 25
5.2 Evaluation of LSTM Model Performance 26
5.3 GARCH-LSTM Ensemble 28
5.4 VaR and ES 29

6 Conclusion **32**

References **35**

1 Introduction

The accurate prediction of financial market volatility has long been a research focus, especially given its implications for risk management, portfolio allocation, and derivative pricing. Volatility is a proxy for uncertainty in asset returns and is central to estimating market risk indicators such as Value-at-Risk (VaR) and Expected Shortfall (ES). Traditional approaches, particularly GARCH-type models introduced by Bollerslev [1986], have become widely adopted because of their interpretability and empirical success in modeling volatility clustering. However, in recent years, deep learning methods have started to create serious competition for classical econometric models in time-series forecasting. One of the examples of such deep learning networks that have been incredibly effective in learning long-range temporal dependencies is a variant of recurrent neural networks called Long-Short-Term Memory (LSTM) network. While the benefit of GARCH models is that they exhibit better statistical interpretability, they could fail to capture complex nonlinear structures in the data. Conversely, LSTM models, while powerful, may be prone to overfitting and often lack the interpretability offered by traditional econometric models (Borovykh et al. [2017]).

In light of these considerations, hybrid modeling techniques have received increasing attention, as they aim to leverage the complementary strengths of both statistical and machine learning approaches. Studies such as Kim and Won [2018] demonstrate that combining GARCH-type models with LSTM networks in integrated framework can reduce forecast errors and improve robustness. However, more research is needed to explore how such ensemble models perform in volatility forecasting and the estimation of VaR and ES, especially when dynamic weighting is applied based on model performance.

This thesis contributes to this effort by comparing the performance of three models on S&P 500 return data spanning two decades: a classical GARCH (1,1), a data-driven LSTM, and a dynamically weighted ensemble of the two. The goal of the research is to empirically test whether the ensemble consistently improves forecast accuracy and risk estimation. Volatility is proxied by squared returns, and all models are evaluated for their ability to minimize forecast error and pass backtests for VaR and ES. The empirical setup begins with rolling window GARCH forecasts with a 252-day window and 20-day refitting, starting in 2005. LSTM is trained using a walk-forward approach with a training horizon of 1008 days and tested on 504-day out-of-sample windows. The ensemble weights are recalculated in each window based on the inverse mean absolute error (MAE) from previous forecasts, allowing the model to adapt over time. All forecasts are compared using error metrics, including RMSE, MAE, and heteroscedastic MSE, and VaR/ES estimates are subjected to Kupiec and Christoffersen backtests.

The main research questions of this research are:

RQ 1: Does the Long-Short-Term-Memory (LSTM) network perform better than classical GARCH(1,1) according to the diagnostic tests?

RQ 2: Does a dynamically weighted ensemble model of GARCH and LSTM outperform standalone models based on chosen metrics?

RQ 3: Does the ensemble model produce the closest number of Value-at-Risk (VaR) violations to the assumed level of α among the three models?

The objective of this research is to evaluate whether a dynamically weighted ensemble of GARCH(1,1) and LSTM models can outperform each individual model in forecasting volatility and estimating Value-at-Risk (VaR) and Expected Shortfall (ES) for S&P 500 returns, thereby

contributing to broader efforts to improve the accuracy of financial risk prediction.

The rest of the thesis is structured as follows. In section 2 literature review with theoretical background is reviewed, as well as the evolution of the volatility prediction methods covering both traditional econometric techniques along with machine learning approaches. The dataset description, data preprocessing as well as feature engineering and volatility proxy are covered in section 3. Subsequently, the methodology and modelling framework are covered in section 4. Afterwards, in section 5, empirical results and error analysis, as well as statistical tests, are discussed. Finally, conclusions, implications along with limitations and directions for future research are presented in the concluding section 6, finishing with references outlining the sources of literature and tools used.

With the research questions and motivation clarified, the following section provides a structured overview of the academic literature on volatility modeling and the estimation of Value-at-Risk (VaR) and Expected Shortfall (ES), beginning with foundational econometric models and progressing toward modern deep learning techniques and integrated approaches.

2 Literature Review

Volatility is defined as a measure of variability or uncertainty in financial time series, often modeled as the conditional variance of asset returns (Tsay [2005]). The volatility modeling of financial time series has evolved significantly over last decades. In early years of financial econometric research there was the assumption of asset return variance being homoscedastic meaning constant over time. Later on the assumption about this condition was refuted due to the inability to explain the key empirical property of financial time series called volatility clustering, which refers to the phenomena of sizable changes in returns being followed by sizable changes, and small changes by small ones. In order to account for this feature the Autoregressive Conditional Heteroskedasticity (ARCH) model was introduced by Engle [1982] which was breaking point in time series econometrics. The ARCH model improved volatility modeling by allowing the conditional variance of returns to depend on past squared residuals, capturing the effect of recent shocks on current volatility.

While ARCH captured short-term memory in volatility, it required a large number of parameters for high persistence, which limited its practicality in applied settings. This limitation led to the development of the Generalized ARCH (GARCH) model by Bollerslev [1986]. The GARCH(1,1) specification became the standard for volatility modeling due to its simplicity and strong empirical performance. In this model, the conditional variance at time t is a weighted average of a long-run variance level, the squared return shock from the previous period, and the previous period's forecast variance.

The variation above experienced widespread adoption for couple of reasons, namely it led to simplification of volatility estimation since it contained lower number of parameters resulting in more parsmonious model as well as showing consistency in performance across a range of financial assets (look Bollerslev et al. [1992], Nelson [1991]). A balance between flexibility and interpretability served as one of the main factors for prolonged popularity of the GARCH model making it suitable for modeling asset return volatility across various markets and time horizons (Andersen et al. [2003]).

However, despite its practical strengths, the model is based on assumptions that often do not reflect real-world market behavior. For instance, it treats positive and negative return shocks

symmetrically, implying that both have the same effect on volatility, a notion that contradicts empirical findings in financial markets, where negative shocks tend to cause disproportionately larger increases in volatility. This phenomenon, known as the leverage effect, has been well documented in the literature (Black [1976], Glosten et al. [1993]). To address these limitations, various asymmetric and nonlinear extensions have been proposed. The Exponential GARCH (EGARCH) model introduced by Nelson [1991] accounts for asymmetries without requiring non-negativity constraints on model parameters. Similarly, the GJR-GARCH model by Glosten et al. [1993] adds an indicator function to allow the model to respond differently to negative returns.

Despite these advances, all GARCH-type models share a common structure based on a parametric functional form, which may limit their ability to adapt to changing market regimes or nonlinear dynamics (Taylor [2008]). Moreover, GARCH models often rely on distributional assumptions such as normal or t-Student. These assumptions can be problematic given the empirical presence of fat tails and skewness in financial returns. To accommodate this, researchers have explored alternative distributions and even non-parametric methods Ardia and Hoogerheide [2014] conducted an empirical evaluation of GARCH and GJR-GARCH models applied to the S&P 500 index, focusing on their accuracy in forecasting Value-at-Risk (VaR) and Expected Shortfall (ES) under different parameter update frequencies. Their findings suggest that while forecast accuracy remains relatively stable across various update intervals, the reliability of risk measures can deteriorate if model assumptions are misaligned with the underlying data-generating process, particularly during periods of heightened market uncertainty. Despite their continued relevance, the limitations of GARCH-type models have prompted researchers to explore more adaptive and non-linear forecasting frameworks. This has led to a growing body of literature applying machine learning techniques, particularly recurrent neural networks (RNNs) and Long Short-Term Memory (LSTM) architectures, to volatility modeling tasks (look Kim and Won [2018], Zhang et al. [2022]).

Because of their potential to detect nonlinearities and unconventional trends that econometric models are too limited to uncover deep learning models have increased in popularity in financial time series modeling in recent years (Gamboa [2017], Sezer et al. [2020]). Among these techniques, recurrent neural networks (RNNs) were among the first designed specifically to handle sequential input. Because RNNs retain a form of memory by integrating information from previous time steps into later computations, they are conceptually well-suited to tasks like time series forecasting and volatility prediction. One of the earliest applications of RNNs in the area of financial forecasting could be observed in research paper done by Kaastra and Boyd [1996] where RNN models were used to predict financial and economic time series. Although no single asset was emphasized, the study provided a comprehensive blueprint for designing neural networks suited to financial data, laying groundwork for subsequent research in data-driven volatility forecasting.

Despite their conceptual suitability for sequential data, standard RNNs face practical limitations namely the vanishing and exploding gradient problem which hinders their ability to learn long-term dependencies in financial time series (Bengio et al. [1994]). As the attempt to go around those limitations the Long Short-Term Memory (LSTM) network, a specialized type of RNN was presented by Hochreiter and Schmidhuber [1997] that was designed to better handle long sequences of data. Incorporation of memory cells and gating mechanisms namely input, forget and output gates, enabled LSTM networks to retain important information over long periods and filter out irrelevant signals which made them great for complex time series such as stock return volatility.

LSTM is not driven by prior assumptions about the data distribution or generation as it is with GARCH or other classical econometric models. Sn contrast, the model learns patterns directly from the data, allowing it to capture nonlinear, nonstationary, and regime-dependent dynamics commonly observed in financial markets. Those properties make the LSTM technique incredibly flexible and well-suited approach for tasks where volatility evolves under structural breaks or shows delayed reactions to past information.

LSTM consistent performance in the context of volatility prediction is widely supported by empirical studies. One of them is study by Yu and Li [2018] which illustrated that LSTM model produced better volatility predictions than forecasts performed by traditional techniques, showing lower forecast errors in terms of both MAE and RMSE. In another paper by Liu [2019] LSTM, GARCH(1,1), and support vector regression (SVR) were compared for forecasting the volatility of the S&P 500 and Apple Inc, taking squared returns as a realized volatility proxy and training the models with 70/30 in and out of sample split, it concluded that LSTM consistently produced the most accurate results achieving lowest MAE and RMSE across both assets.

However, LSTM networks as most deep learning networks require relatively large sample size and careful hyperparameter tuning to learn effectively. Moreover, another drawback of LSTMs is their black box nature which raises problems related to interpretability which is highly important topic in finance and risk modelling since transparency is paramount for meeting regulatory requirements (look Hosseini and Kiani [2022]).

With the growing popularity of LSTMs and other deep learning techniques the natural progression of research was to explore various ways to enhance the model performance. One of the ideas was to combine LSTM and GARCH approaches into integrated model, one of the first frameworks for this was introduced by Kim and Won [2018]. Their core hypothesis was that the out of sample forecast accuracy could me improved by feeding structured volatility estimates, derived from GARCH and other statistical models, into the LSTM network. Empirical results based on the KOSPI 200 index shown that there is not reason to reject this hypothesis, as the integrated models consistently outperformed both standalone LSTM and traditional models across several error metrics, including MAE, MSE, HMAE, and HMSE.

However there are other frameworks for combining two or more models one of the approaches is called weighted ensemble. While hybrid models like those of Kim and Won [2018] integrate statistical features into a single predictive model, ensemble approaches combine the outputs of independently trained models. In weighted ensemble, each model is trained separately and produces its own forecast, which is then aggregated at the output level based on predefined or adaptive weighting schemes. Due to this modular design, each model has the opportunity to contribute using its own modeling strengths, for example GARCH provides a statistically sound and interpretable framework to capture volatility persistence and mean-reversion in time series, while LSTM is well-rounded to learn complex, nonlinear and potentially regime-dependent temporal patterns and in ensemble all of this could be performed in more dynamic nature.

One of the first applications of such ensemble time series modeling with a performance-based integration methodology is that of Kristjanpoller and Minutolo [2016], who suggested a weighting between GARCH and artificial neural networks (ANN). In their model, the final forecast of crude oil prices was produced using weighted average of the outputs of the two models, with the weights being based on their recent predictive errors. Although this confirmed the practicality of adaptive ensemble methods, the model was constrained by static architectures and lacked deep sequence modeling abilities that of LSTM.

In contrast, the contribution of this thesis lies in developing a dynamically weighted ensemble model that combines forecasts from a classical GARCH(1,1) model and a separately trained LSTM network. Rather than fusing inputs or using static weights, the two models generate independent volatility forecasts, which are then combined using performance-based dynamic weighting. Specifically, weights are computed based on the inverse of each model's Mean Absolute Error (MAE) over two rolling historical windows, enabling the ensemble to adapt over time based on recent forecasting accuracy. This method balances the interpretability and statistical rigor of GARCH models with the nonlinear learning and adaptability of LSTM networks. It also introduces temporal adaptability and forecast-driven integration into the ensemble design, offering a novel contribution to volatility forecasting literature by allowing the model combination to respond directly to changing market dynamics.

Forecasting financial volatility is essential for effective risk management, particularly in estimating Value-at-Risk (VaR) and Expected Shortfall (ES). These measures are widely used by financial institutions to quantify potential losses under adverse conditions and are central to regulatory frameworks such as Basel III, which now prioritizes ES over VaR due to its improved sensitivity to tail risk (look on Banking Supervision [2016]).

VaR estimates the maximum expected loss over a given time horizon at a specified confidence level. For example, a 1-day 95% VaR of $1 million implies there is a 5% chance of losing more than that amount in a day. However, VaR does not account for the magnitude of losses beyond this threshold. ES, also known as Conditional VaR, addresses this limitation by estimating the average loss in the worst $(1-\alpha)\%$ of cases, offering a more comprehensive view of downside risk. Both VaR and ES depend on accurate one-step-ahead forecasts of conditional volatility (McNeil et al. [2015]). The reliability of these risk forecasts must be assessed using formal backtesting procedures. Kupiec [1995] proposed the unconditional coverage test to evaluate whether the frequency of VaR exceedances aligns with the expected level. Christoffersen [1998] extended this by testing whether exceedances occur independently over time.

While much research has focused on improving volatility forecasts through integrated models, such gains do not always translate into more accurate risk estimates. **?** proposed a hybrid GARCH-GRU model, using GARCH(1,1) volatility forecasts as input to a GRU network. When tested on the S&P 500 index, the model achieved lower mean squared error and log loss in volatility prediction compared to standalone GARCH. However, these improvements were not consistently reflected in risk forecasting performance. In backtests of VaR and ES under both normal and Student-t distributions, the hybrid model showed no clear advantage over GARCH in unconditional or conditional coverage tests. These findings reflect a broader insight in the literature: improvements in point forecast accuracy for volatility do not necessarily imply better performance in estimating tail risk. As research increasingly incorporates complex models, the distinction between predictive accuracy and risk reliability has become more clear underscoring the need to evaluate models not only on average error metrics, but also on their effectiveness in capturing extreme outcomes under formal backtesting procedures.

As a contribution to this evolving body of research, this thesis proposes a novel dynamically weighted ensemble model that integrates the structured interpretability of GARCH(1,1) with the nonlinear learning capabilities of LSTM networks. By adjusting model weights based on recent forecasting performance, the ensemble approach aims to enhance adaptability across different market regimes. Thus, this study contributes to the literature by empirically evaluating whether such a performance-based ensemble can improve both volatility forecasting and the estimation of Value-at-Risk (VaR) and Expected Shortfall (ES), relative to its individual components, using

S&P 500 return data.

3 Dataset description

Building on the literature review, this section describes the data sources, preprocessing steps, and key features used in the volatility forecasting models. Figures 1 and 2 show the computed log returns, which are based on daily data from the S&P 500 index comprising 5,032 observations spanning the period from January 1, 2004, to January 1, 2024. The dataset was obtained from Yahoo Finance, with daily closing prices used to compute logarithmic returns, as shown in Equation 1. Log returns are preferred in financial time series analysis for their ability to normalize data and improve the consistency and comparability of price changes over time. Figures 3 and 4 illustrate the changes in realized volatility, which was derived to capture the historical variability in returns and serves as the target variable, providing a benchmark for model forecasting performance. The derivation of realized volatility is presented in Equation 2. The log returns are subsequently employed as predictors of future conditional volatility within the GARCH model. Conditional volatility forecasts generated by the GARCH model are then combined with predictions from a Long Short-Term Memory (LSTM) neural network, which is trained on absolute returns. This integration is achieved through a dynamic weighting scheme that allows the model to adjust the relative influence of each component over time. The resulting hybrid framework aims to more effectively represent the complex and evolving nature of volatility in the S&P 500 index.

Statistic	**Close Prices**	**Log Return**
Count	5033	5032
Mean	2139.0915	0.0003
Standard Deviation	1085.5978	0.0121
Minimum	676.5300	-0.1277
25% Quantile	1267.3700	-0.0041
50% Quantile (Median)	1805.4500	0.0007
75% Quantile	2798.0950	0.0056
Maximum	4796.5601	0.1096

Table 1: Descriptive Statistics of S&P 500 Close Prices and Log Returns. Source: Yahoo Finance

Table 1 presents key descriptive statistics for the daily closing prices and logarithmic returns of the S&P 500 index over the period from 2004 to 2024. The average daily closing price was approximately 2,139, while the median stood at 1,805, indicating a right-skewed distribution and suggesting that a greater number of observations fell below the mean. In contrast, the log returns exhibit a more symmetric distribution. The mean return was 0.0003, and the median was 0.0007, implying that the returns are relatively balanced and consistent with a distribution that approximates normality. The standard deviation of closing prices was 1,085, reflecting substantial variability in market levels over the two-decade period. For log returns, the standard deviation was 0.0121, indicating moderate daily fluctuations. Notably, the most extreme daily loss was -12.77%, while the largest daily gain reached 10.96%, underscoring the presence of significant market shocks during the sample period. The 25th, 50th, and 75th percentiles of the log returns were -0.0041, 0.0007, and 0.0056, respectively, highlighting that while most daily returns were relatively modest, they exhibited variability in both directions. Collectively, these

summary statistics provide a comprehensive snapshot of the market's behavior, capturing both central tendencies and the extent of volatility across the 20-year horizon.

The daily logarithmic returns of the S&P 500 index used in this study are computed by taking the natural logarithm of the ratio of consecutive daily closing prices. Specifically, the return at day t is calculated as shown in Equation 1:

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right) \tag{1}$$

where P_t denotes the closing price of the index at time t, and r_t represents the corresponding log return.

To complement the summary statistics presented in Table 1, Figures 1 and Figure 2 provide visual representations of the logarithmic returns derived from 20 years of S&P 500 daily data. Figure 1 displays the density plot of log returns, offering insights into the overall distributional characteristics, such as symmetry, kurtosis, and potential skewness. Figure 2 illustrates the change of log returns over time, highlighting periods of heightened and subdued market activity across the sample window.

The density plot in Figure 1 illustrates that the empirical distribution of log returns (Equation 1) deviates from normality, showing a sharper peak and heavier tails. This leptokurtic pattern suggests the presence of excess kurtosis and extreme return values, which are characteristic of financial time series and may reflect market shocks or structural breaks. Together, these visualizations enhance the interpretation of the descriptive statistics and provide a clearer understanding of log returns dynamics over the observed period.

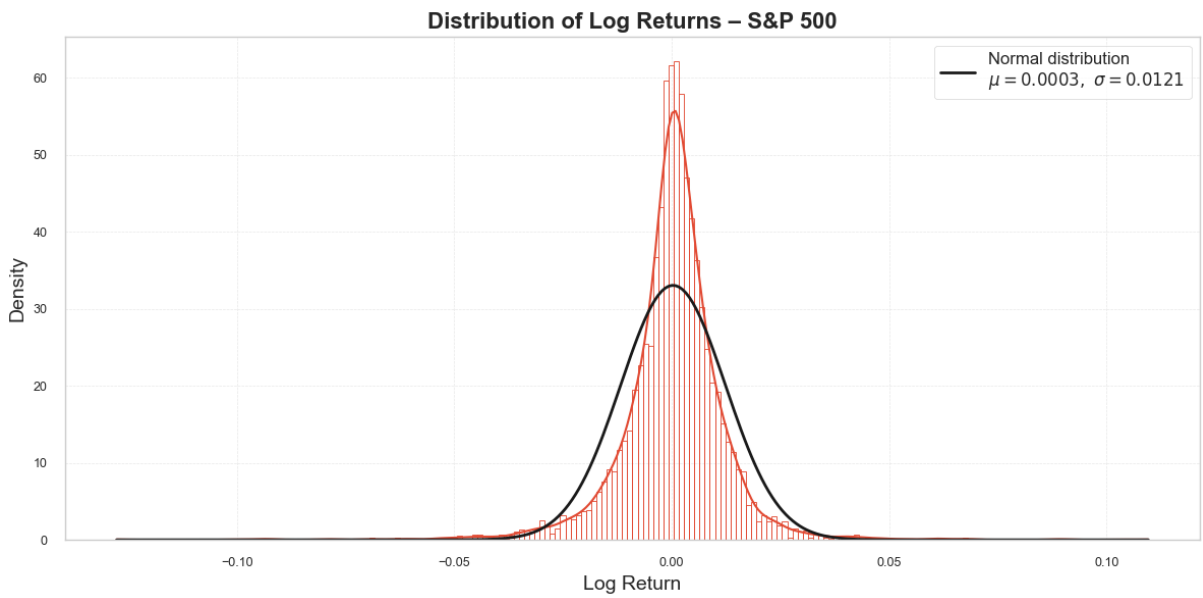

Figure 1: Comparison of the S&P 500 log return distribution (from Jan 1 2024 to Jan 1 2024) with a normal distribution. Source: Yahoo Finance

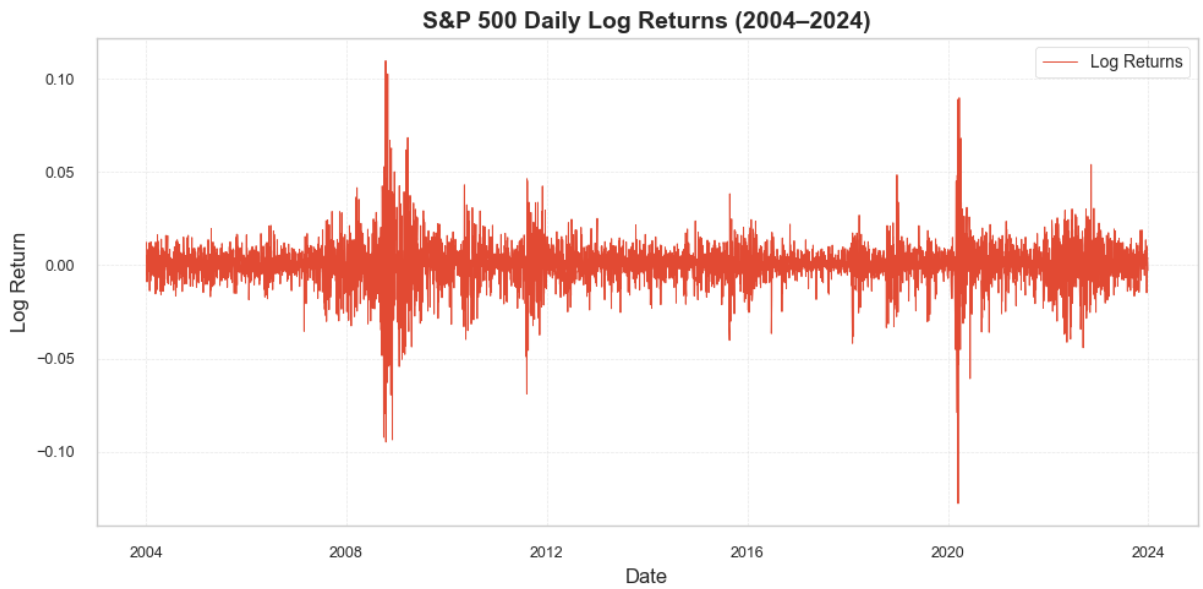

Figure 2: Daily log returns of the S&P 500 from 2004–01–01 to 2024–01–01. Source: Yahoo Finance

The daily realized volatility in this paper is estimated using squared logarithmic returns as a proxy, a widely used and straightforward method for capturing the magnitude of daily market fluctuations. The realized volatility on day t, denoted as σ_t^2, is computed as shown in Equation 2.

$$\sigma_t^2 = r_t^2 \tag{2}$$

where r_t represents the log return on day t. Squaring the return eliminates the direction of price changes, thereby isolating the magnitude of market movements. This measure reflects the extent of variability in returns and serves as a proxy for the intensity of market activity or turbulence on a given day.

Figures 3 and 4 provide visual representations of the daily realized volatility derived from 20 years of S&P 500 data using Equations 1 and 2. Figure 3 displays the distribution of realized daily volatility, overlaid with a normal density function. The plot highlights the skewness and leptokurtosis of the data, indicative of the presence of extreme events, which aligns with the findings from the descriptive statistics. Figure 4, on the other hand, illustrates the time series of daily realized volatility of S&P index over 20 years, showcasing periods of heightened volatility, especially around the global financial crisis in 2008 and the COVID-19 market reaction in 2020.

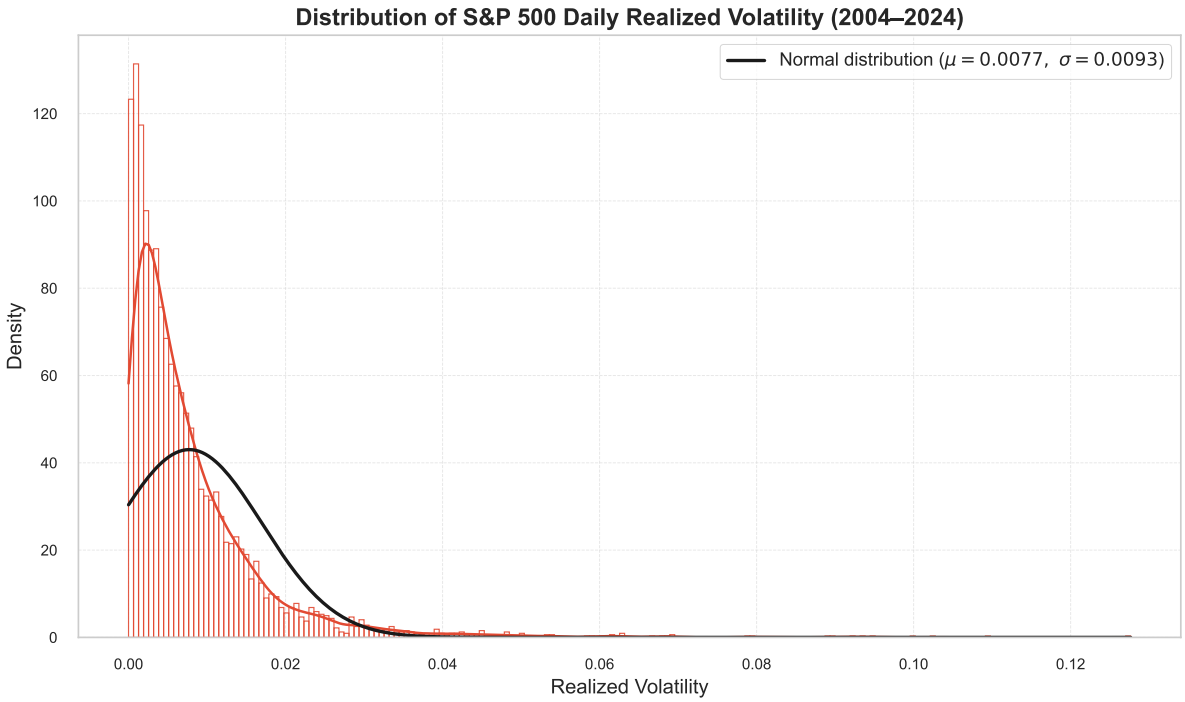

Figure 3: Distribution of daily volatility (absolute log returns) in the S&P 500 from 2004–01–01 to 2024–01–01, overlaid with a fitted normal distribution. Source: Yahoo Finance

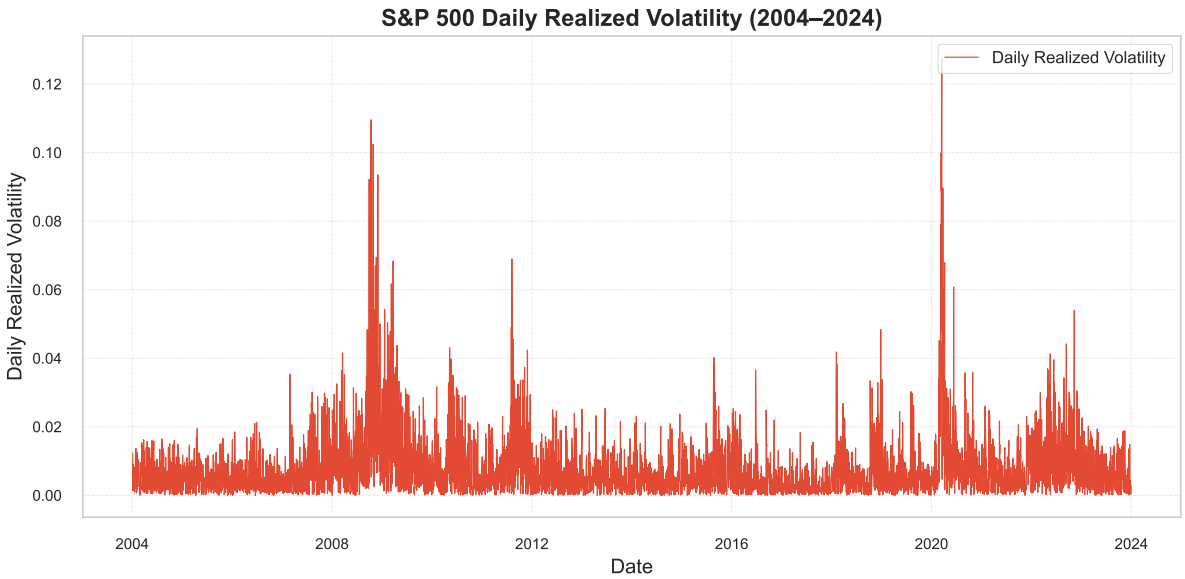

Figure 4: Daily volatility (absolute log returns) of the S&P 500 from 2004–01–01 to 2024–01–01. Source: Yahoo Finance

4 Methodology and Empirical Framework

This study integrates machine learning techniques with traditional econometric time series models to forecast the daily realized volatility of the S&P 500 index and to estimate both Value

at Risk (VaR) and Expected Shortfall (ES). The modeling framework draws upon two distinct, yet complementary approaches: a GARCH (Generalized Autoregressive Conditional Heteroskedasticity) model and a Long Short-Term Memory (LSTM) neural network. Logarithmic returns, which are widely regarded for their strong predictive power in volatility modeling, serve as the primary input for the GARCH model, providing insights into future conditional volatility. In parallel, absolute returns are employed as inputs for the LSTM network, leveraging the model's capacity to capture complex nonlinear patterns and temporal dependencies in financial time series data. This chapter provides a detailed overview of the methodology applied in this study, focusing on the implementation of the GARCH, LSTM, and their ensemble models, as well as the approach used for the estimation of Value at Risk (VaR) and Expected Shortfall (ES) based on the realized volatility predictions of those models. The discussion builds upon the empirical dataset described in Section 3 and outlines the modeling procedures, forecast strategies, and evaluation techniques employed to assess volatility and risk dynamics.

4.1 GARCH

Bollerslev's GARCH (Generalized Autoregressive Conditional Heteroskedasticity) model, an extension of the ARCH (Autoregressive Conditional Heteroskedasticity) framework, models the conditional variance as a weighted sum of past squared residuals and past conditional variances. This structure allows the GARCH model to effectively capture volatility clustering, a common phenomenon in financial time series where periods of high volatility tend to be followed by high volatility, and low volatility by low volatility.

4.1.1 Model Choice and Configuration

As outlined by **?**, the GARCH model specifies the conditional variance at time t, denoted as σ_t^2, as a function of both past squared residuals (errors) and past conditional variances. The parameters q and p represent the orders of the ARCH and GARCH components, respectively, determining how many lagged squared shocks and past volatility terms are incorporated in the model. This recursive structure enables the model to dynamically adjust volatility forecasts in response to new information, capturing key features of financial time series such as volatility persistence and clustering.

Mathematically, the GARCH(p, q) model is defined as shown in Equation 3.

$$\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2 \tag{3}$$

where:

- $\omega > 0$ is the constant term,
- α_i are the ARCH parameters measuring the impact of past shocks (short-term volatility),
- β_j are the GARCH parameters capturing the persistence of volatility over time,
- ε_{t-i} are the lagged residuals,
- σ_{t-j}^2 are the lagged conditional variances.

A large sum of α_i and β_j parameters close to one in Equation 3 indicates that volatility shocks are long-lasting, which is consistent with the persistence typically observed in financial time series data.

To illustrate a more compact representation, the model can also be expressed using lag polynomials as it is illustrated in Equation 4:

$$\sigma_t^2 = \omega + \alpha(B)\varepsilon_t^2 + \beta(B)\sigma_t^2 \tag{4}$$

where the polynomial lag operators are defined as in Equations 5 and 6:

$$\alpha(B) = \sum_{i=1}^{q} \alpha_i B^i \tag{5}$$

$$\beta(B) = \sum_{j=1}^{p} \beta_j B^j \tag{6}$$

and B is the backshift operator such that $B^k x_t = x_{t-k}$. When all $\beta_j = 0$, the GARCH model reduces to the ARCH(q) process, relying solely on recent squared residuals.

To determine the optimal configuration of the GARCH(p, q) model, an automated algorithm was implemented to evaluate all combinations of p and q parameters ranging from 0 to 2 using the S&P 500 log returns. Each model specification was assessed based on the Bayesian Information Criterion (BIC) and the Akaike Information Criterion (AIC), with the lowest values indicating the best-fitting model. These criteria are defined in Equations 7 and 8 as follows:

$$\text{AIC} = -2\ln(\hat{L}) + 2k \tag{7}$$

$$\text{BIC} = -2\ln(\hat{L}) + k\ln(n) \tag{8}$$

where $\hat{L}$ is the maximum likelihood of the model, k is the number of estimated parameters, and n is the number of observations. Both criteria penalize model complexity, but BIC applies a stronger penalty for additional parameters, favoring more parsimonious models.

Following this selection procedure, the GARCH(1,2) model was identified as the preferred specification, as it achieved the lowest AIC and BIC values among the evaluated candidates. To further assess model adequacy, the Lagrange Multiplier ARCH (LM-ARCH) test was applied to the standardized residuals of both the benchmark GARCH(1,1) model and the selected GARCH(1,2) model. This diagnostic test evaluates the presence of remaining ARCH effects in the residuals, which would suggest that conditional heteroskedasticity has not been fully captured.

The LM-ARCH test yielded high p-values for both models (0.9979 in each case), indicating no statistically significant autocorrelation in the squared residuals. Thus, both specifications appear to adequately account for the time-varying volatility in the return series.

Given the similar diagnostic performance and guided by the principle of parsimony, the GARCH(1,1) model was ultimately chosen. Despite its marginally higher AIC and BIC values (Equations 7 and 8) it offers a more interpretable and computationally efficient structure without compromising statistical adequacy.

4.1.2 GARCH Realized Volatility Prediction Approach

The GARCH(1,1) model (Equation 3) is employed to generate one-step-ahead forecasts of S&P 500 volatility using a rolling window approach. A window size of 252 daily log return observations, approximately equivalent to one trading year, is used, with the model refitted every 14 days. These choices are consistent with common practice in the financial econometrics literature, where the 252-day window reflects a typical annual trading cycle, and the 14-day refit interval is frequently adopted as a practical compromise between model adaptability and computational cost.

Prior to model estimation, the stationarity of the log return series was verified using the Augmented Dickey-Fuller (ADF) test. The test results provided strong evidence against the presence of a unit root, confirming that the log return series is stationary which is a necessary condition for the application of GARCH-type models.

The forecasting procedure begins by estimating the GARCH(1,1) model using the first 252 S&P 500 daily log returns, corresponding to one trading year of data starting from January 1, 2004. Based on this initial estimation window, the model generates a one-step-ahead forecast for the conditional volatility on day $t = 253$. Using the same fitted model, volatility forecasts are then produced for the next 14 days. After this forecasting period, the rolling window is shifted by 14 days, and the model is re-estimated using the most recent 252 daily returns. This re-estimation and forecasting cycle is repeated throughout the sample period, ending with the final forecast based on data available up to January 1, 2024.

This rolling estimation framework allows the model to adapt to evolving market conditions while maintaining a consistent window length for parameter estimation.

4.2 LSTM

The Long Short-Term Memory (LSTM) network is an advanced type of recurrent neural network (RNN), specifically designed to overcome the limitations of standard RNNs in modeling long-term dependencies. While traditional RNNs are structured to learn sequential patterns, they suffer from the problem of vanishing and exploding gradients, which severely hinders their ability to retain information over long sequences. In response to these challenges, Hochreiter and Schmidhuber [1997] introduced the LSTM architecture. The key innovation of LSTM networks lies in the introduction of memory cells and gating mechanisms, specifically, the input gate, forget gate, and output gate that regulate the flow of information over time. These gates enable the model to selectively retain or discard information at each time step, thereby preserving relevant long-term dependencies while filtering out irrelevant data. In simpler terms, the LSTM architecture acts as a dynamic data filter, allowing the network to prioritize information with high relevance and maintain contextual knowledge across extended sequences. This ability makes LSTM networks particularly effective for modeling both short-term fluctuations and long-term trends in sequential data, such as financial time series.

4.2.1 LSTM Architecture

The operation of the LSTM network is illustrated in Figure 5.

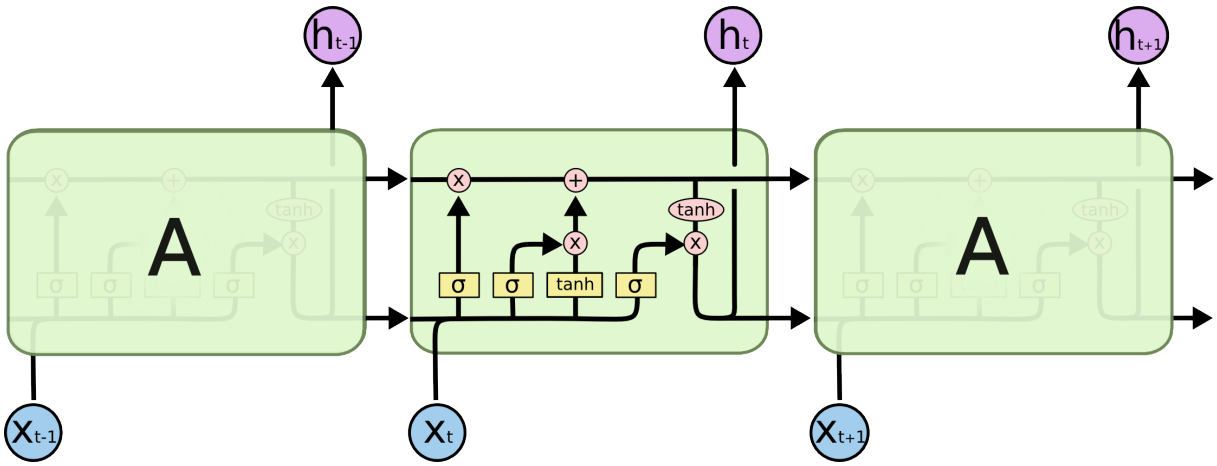

Figure 5: Diagram of an LSTM cell showing the information flow through the forget, input, and output gates. Source: adapted from Olah (2015)

At each time step t, the LSTM processes two inputs: the current data input x_t and the hidden state from the previous time step h_{t-1}. Based on these inputs, the LSTM first computes the forget gate activation, defined in Equation 9:

$$f_t = \sigma(U_f x_t + V_f h_{t-1} + b_f), \tag{9}$$

where U_f and V_f are the weight matrices associated with the input and the hidden state respectively, b_f is the bias vector, and $\sigma(\cdot)$ denotes the sigmoid activation function. The output f_t is a vector with elements in the range $(0, 1)$, controlling the extent to which information from the previous cell state C_{t-1} is retained. This gating results in an intermediate updated cell state shown in Equation 10:

$$C'_t = f_t \odot C_{t-1}, \tag{10}$$

where $\odot$ represents element-wise multiplication.

Next, the LSTM evaluates the input gate, which determines the amount of new information to incorporate into the cell state. The activation of the input gate is defined in Equation 11:

$$i_t = \sigma(U_i x_t + V_i h_{t-1} + b_i), \tag{11}$$

where U_i, V_i, and b_i are the corresponding input weight matrix, hidden state weight matrix, and bias vector for the input gate. At the same time, the candidate cell state $\tilde{C}_t$ is generated by Equation 12:

$$\tilde{C}_t = \tanh(U_c x_t + V_c h_{t-1} + b_c), \tag{12}$$

where U_c and V_c are the weight matrices for the candidate generation and b_c is the bias term. The hyperbolic tangent function $\tanh(\cdot)$ ensures that the candidate values are within the range $(-1, 1)$.

The cell state is then updated by combining the retained information and the new candidate state as in Equation 13:

$$C_t = C'_t + i_t \odot \tilde{C}_t, \tag{13}$$

thus blending the old and new information according to the learned gates.

The output gate controls the flow of information from the cell state to the hidden state. It is computed using Equation 14:

$$o_t = \sigma(U_o x_t + V_o h_{t-1} + b_o), \tag{14}$$

where U_o, V_o, and b_o are the respective input weights, hidden state weights, and bias term for the output gate. Finally, the hidden state is updated by Equation 15:

$$h_t = o_t \odot \tanh(C_t), \tag{15}$$

where h_t serves both as the output of the LSTM cell for the current time step and as the input hidden state for the next time step.

This gated structure (Figure 5) allows the LSTM to effectively regulate the storage, update, and retrieval of information across long sequences. By learning to adjust the behavior of its gates during training, the LSTM can capture complex temporal dependencies and maintain relevant information over extended time horizons. This capability makes LSTM networks particularly powerful for tasks such as time series forecasting (**?**).

4.2.2 Hyperparameter Tuning and Model Setup

To improve the accuracy of the neural network's realized volatility forecasts, an extensive hyperparameter tuning process was conducted using 1,008 days of empirical log return data. The Random Search algorithm, implemented via the KerasTuner (Bergstra and Bengio [2012]) library in Python, was utilized to identify the optimal combination of hyperparameters, with model performance evaluated based on the Mean Squared Error (MSE) loss function. Random Search operates by randomly sampling combinations of hyperparameters from predefined search spaces and evaluating each configuration's performance on a validation set. The combination that yields the lowest MSE on the validation data is selected as the optimal set of hyperparameters. This approach offers a computationally efficient alternative to exhaustive grid search.

The hyperparameter tuning process explored the following search spaces: the number of units in each of the three LSTM layers (64, 128, 256, 512), the activation function (tanh, ReLU, ELU), the learning rate for the Adam optimizer (0.01, 0.005, 0.0009, 0.00015), the dropout rate (0.1, 0.4, 0.01, 0.0002, 0.05), and the L2 regularization coefficient (0.00009, 0.00004, 0.00002, 0.00012).

Following an extensive Random Search procedure, the optimal LSTM architecture was selected (Table 2). The final model consisted of three stacked LSTM layers with 128, 512, and 128 units, respectively, each employing an L2 regularization parameter of 0.0001 and the tanh activation function. Each LSTM layer was followed by a dropout layer with a dropout rate of 0.0005 to mitigate overfitting. The network concluded with a dense output layer comprising a

single neuron and a linear activation function. The Adam optimizer was used with a learning rate of 0.01. This hyperparameter configuration was held fixed across the entire forecasting period to produce one-step-ahead forecasts of realized volatility.

Component	**Configuration**
Number of LSTM layers	3
Units per LSTM layer	128, 512, 128
Activation function (LSTM layers)	tanh
L2 regularization coefficient	0.0001
Dropout rate (after each LSTM layer)	0.0005
Optimizer	Adam
Learning rate	0.01
Output layer	Dense (1 neuron)
Activation function (output)	Linear
Loss function	Mean Squared Error (MSE)

Table 2: Final LSTM network architecture and hyperparameter settings.

4.2.3 LSTM Forecasting Approach

The LSTM model, configured with the hyperparameters described in the preceding subsection and illustrated in Table 2, was employed to forecast realized volatility. The dataset constructed for this task consisted of absolute values of daily log returns and lagged realized volatility observations, where realized volatility served as the target variable.

The input features, namely the log returns, were organized into sequences of three consecutive observations, a choice made based on an arbitrary but practically reasonable decision to allow the model to capture short-term temporal dependencies. For each input sequence, the corresponding target was the realized volatility of the subsequent trading day, enabling the LSTM network to learn the relationship between recent return dynamics and future volatility levels. The model was trained using the Adam optimizer and employed the Mean Squared Error (MSE) loss function, both of which are standard choices for regression tasks involving continuous outputs. This setup enabled the LSTM to effectively model the nonlinear and dynamic relationship between returns and volatility in the S&P 500 index.

The walk-forward methodology was employed (Figure 6) to train the LSTM model and generate forecasts for one-step-ahead realized volatility. The dataset, comprising 4,780 paired observations of daily log returns and realized volatility, was divided into rolling training windows, each consisting of 1,008 observations (approximately four years of data). Within each training window, the data were further split into a training set and a validation set, with the validation set comprising 33% of the window size. Following model training, out-of-sample forecasts were produced using a testing window consisting of the subsequent 500 daily observations. After generating the forecasts, the training window was advanced by 500 observations, and the model was retrained on the new 1,008-day window to produce the next set of 500 out-of-sample predictions. This walk-forward procedure was repeated iteratively until the end of the dataset was reached. All resulting test windows, containing both the predicted and the actual realized volatility values, were concatenated into a single DataFrame for further performance evaluation and analysis.

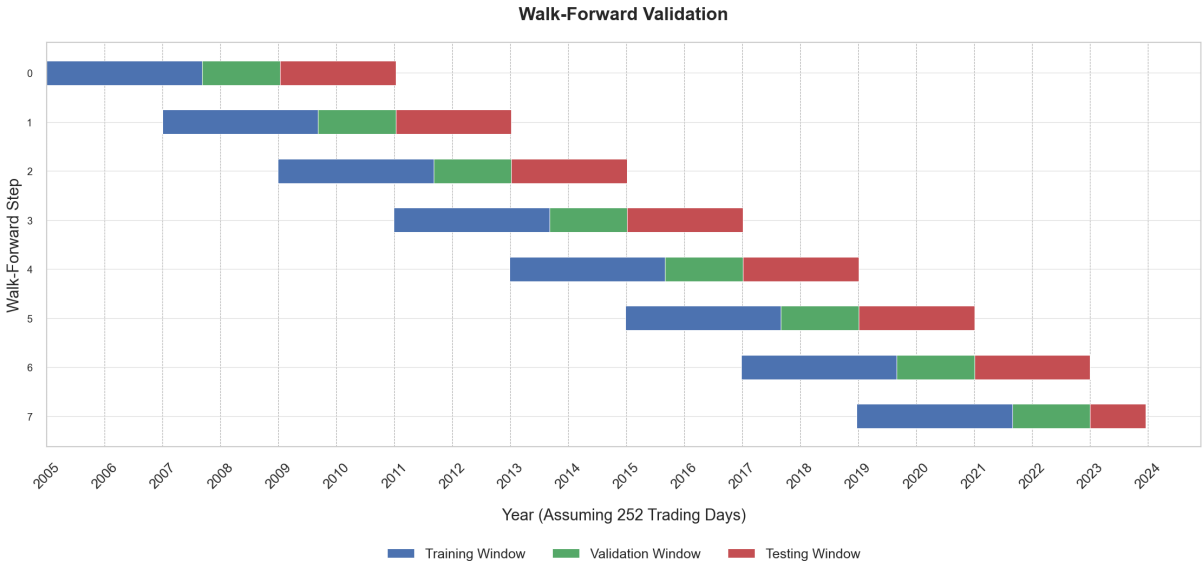

Figure 6: Walk Forward Validation

4.3 GARCH-LSTM Ensemble Model

The GARCH-LSTM hybrid model represents an approach that integrates the strengths of both econometric and machine learning techniques to enhance the accuracy of realized volatility forecasts. This study proposes a novel integration method based on a dynamic weighted ensemble, which adaptively combines the forecasts generated by the GARCH(1,1) model and the LSTM network.

Unlike static ensembling methods that apply fixed weights throughout the forecasting horizon, the dynamic weighted ensemble assigns time-varying weights to the individual models. This allows the hybrid model to adjust the relative importance of the GARCH and LSTM forecasts based on their predictive effectiveness during different market conditions. By dynamically recalibrating model contributions, the ensemble seeks to exploit the complementary properties of GARCH's structured volatility modeling and LSTM's ability to capture nonlinear and long-term dependencies, thereby improving forecast robustness and accuracy.

4.3.1 GARCH-LSTM Forecasting Approach

The ensemble model operates by dynamically combining the predictions generated by the LSTM network and the GARCH(1,1) model. The ensemble forecast at time $t+1$ is computed as a weighted average of the two models' predictions as illustrated in Equation 16:

$$\hat{y}_{t+1}^{\text{ensemble}} = w_k \cdot \hat{y}_{t+1}^{\text{GARCH}} + (1 - w_k) \cdot \hat{y}_{t+1}^{\text{LSTM}} \tag{16}$$

where:

- $\hat{y}_{t+1}^{\text{ensemble}}$ denotes the ensemble forecast of volatility at time $t+1$,
- w_k represents the dynamic weight assigned to the GARCH(1,1) model at the window k,
- $1 - w_k$ represents the dynamic weight assigned to the LSTM model at the window k,
- $\hat{y}_{t+1}^{\text{GARCH}}$ is the predicted volatility from the GARCH(1,1) model at time $t+1$,

- $\hat{y}_{t+1}^{\text{LSTM}}$ is the predicted volatility from the LSTM model at time $t+1$.

Forecasting begins on January 9, 2009, corresponding to the first available LSTM prediction. The forecast horizon is divided into seven windows, each consisting of 500 observations, consistent with the windowing used in the LSTM walk forward approach (Figure 6).

Initially, equal weights of 0.5 are assigned to both models for combining predictions within the first 500-day window. To dynamically update the model weights in subsequent windows, an inverse-error weighting scheme based on the Mean Absolute Error (MAE) is employed. The key idea is to dynamically assign greater weight to the model that has exhibited better forecasting accuracy, measured by achieving a lower Mean Absolute Error (MAE) over the most recent evaluation windows.

First, for the respective window, the MAE is computed separately for the GARCH(1,1) model and the LSTM network using Equations 17 and 18:

$$\text{MAE}_{\text{GARCH}} = \frac{1}{N}\sum_{t=1}^{N}\left|y_t - \hat{y}_t^{\text{GARCH}}\right| \tag{17}$$

$$\text{MAE}_{\text{LSTM}} = \frac{1}{N}\sum_{t=1}^{N}\left|y_t - \hat{y}_t^{\text{LSTM}}\right| \tag{18}$$

where:

- y_t denotes the realized volatility observed at time t,
- $\hat{y}_t^{\text{GARCH}}$ represents the forecasted volatility from the GARCH(1,1) model at time t,
- $\hat{y}_t^{\text{LSTM}}$ represents the forecasted volatility from the LSTM model at time t,
- N is the number of observations within the given window.

Starting from the second window onward, the dynamic model weights are calculated using the average MAE over the two most recent windows. The updated dynamic weights for the LSTM and GARCH models are determined by the Equations 19 and 20:

$$w_{\text{LSTM}} = \frac{\text{Average MAE}_{\text{GARCH}}}{\text{Average MAE}_{\text{LSTM}} + \text{Average MAE}_{\text{GARCH}}} \tag{19}$$

$$w_{\text{GARCH}} = \frac{\text{Average MAE}_{\text{LSTM}}}{\text{Average MAE}_{\text{LSTM}} + \text{Average MAE}_{\text{GARCH}}} \tag{20}$$

where:

- w_{LSTM} is the weight assigned to the LSTM model in the ensemble forecast,
- w_{GARCH} is the weight assigned to the GARCH(1,1) model in the ensemble forecast,
- Average MAE_{LSTM} denotes the mean of the LSTM's MAE values over the previous two windows,
- Average $\text{MAE}_{\text{GARCH}}$ denotes the mean of the GARCH(1,1) model's MAE values over the previous two windows.

Thus, the model with the lower forecasting error (Average MAE) in the previous two periods receives a higher weight in subsequent ensemble forecasts. This procedure ensures that the ensemble dynamically adapts to the relative predictive performance of the GARCH and LSTM models over time, improving the robustness and accuracy of realized volatility predictions.

4.4 VaR and ES

Following the prediction of realized volatility, Value at Risk (VaR) and Expected Shortfall (ES) were estimated to complete the S&P 500 risk modeling framework. Value at Risk (VaR) is widely used in risk management due to its ability to quantify the potential loss in the value of an asset or portfolio over a specified time horizon, given a particular confidence level. It estimates the maximum expected loss that will not be exceeded with a certain probability under normal market conditions. This study uses the volatility forecasts obtained from the GARCH(1,1), LSTM, and ensemble models to compute one-day-ahead VaR estimates. These estimates provide insight into the potential downside risk under normal market conditions and serve as a foundational component of the overall market risk assessment.

Assuming that the standardized residuals of returns are normally distributed, the Value-at-Risk (VaR) [Jorion, 2007] at confidence level α can be computed using Equation 21:

$$\text{VaR}_\alpha = \mu_t + z_\alpha \cdot \sigma_t \tag{21}$$

where:

- VaR_α is the Value at Risk at confidence level α,
- μ_t is the conditional mean (expected return) of the asset at time t, which in this study is assumed to be zero due to the generally negligible magnitude of mean returns compared to volatility,
- z_α is the quantile of the standard normal distribution corresponding to the confidence level α,
- σ_t is the forecasted realized volatility of S&P 500 returns at time t.

For example, for a 95% confidence level, $z_{0.95} \approx 1.645$. The VaR represents the worst expected loss over a given horizon such that the probability of a greater loss is only $1 - \alpha$.

While VaR is useful in quantifying the threshold loss not expected to be exceeded with a given confidence level, it does not account for the magnitude of losses beyond this threshold. To address this limitation, Expected Shortfall (ES), also known as Conditional Value at Risk (CVaR), is employed. ES measures the average loss assuming that the loss has already exceeded the VaR level, thereby providing a more comprehensive assessment of tail risk.

Assuming that the standardized residuals of returns are normally distributed, the Expected Shortfall at confidence level α can be calculated using Equation 22:

$$\text{ES}_\alpha = \mu_t + \frac{\phi(z_\alpha)}{1-\alpha} \cdot \sigma_t \tag{22}$$

where:

- ES_α is the Expected Shortfall at confidence level α,

- μ_t is the conditional mean (expected return) of the asset at time t, assumed to be zero in this study for reasons discussed previously (Equation 21),
- $\phi(z_\alpha)$ is the probability density function (PDF) of the standard normal distribution evaluated at z_α,
- z_α is the standard normal quantile corresponding to the confidence level α, i.e., $z_\alpha = \Phi^{-1}(\alpha)$,
- σ_t is the forecasted volatility of S&P 500 returns at time t.

Expected Shortfall (Equation 22) provides the average of losses that occur in the worst $(1-\alpha)\%$ of cases. For instance, at a 95% confidence level, ES represents the expected loss on days when losses exceed the 1-day 95% VaR. This makes ES a more conservative and robust risk measure, particularly under distributions with fat tails, and it is favored by regulatory frameworks such as Basel III for its coherence and subadditivity properties.

Subsequently, a backtesting procedure is conducted on the VaR and ES estimates to evaluate their accuracy and reliability in capturing downside market risk. In this study, two statistical tests are employed for the evaluation of the VaR forecasts: the Kupiec Proportion of Failures (POF) test and the Christoffersen Conditional Coverage test. The Kupiec test (Kupiec [1995]) assesses whether the number of observed violations namely instances where actual losses exceed the predicted VaR is consistent with the expected violation rate implied by the chosen confidence level. This test focuses solely on the unconditional coverage of the VaR model. It tests whether the correct proportion of breaches occurs over time. The Christoffersen test (Christoffersen [1998]) extends the evaluation by testing for conditional coverage, which includes not only the correct frequency of violations but also the independence of their occurrence. This ensures that VaR breaches are randomly distributed over time, rather than appearing in clusters a pattern that would indicate potential model misspecification or failure to capture time-varying risk. For the ES forecasts, the McNeil and Frey test (McNeil and Frey [2000]) is applied to verify that the average magnitude of losses exceeding the VaR threshold aligns with the expected shortfall estimates, thus providing a comprehensive validation of tail risk predictions. Together, these tests provide a rigorous statistical framework to validate the adequacy of VaR models in a real-world risk management context.

4.5 Model Evaluation Metrics

Error metrics play a crucial role in evaluating the accuracy and overall performance of predictive models. In this study, four key error metrics are employed to assess model effectiveness: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Heteroscedastic Mean Squared Error (HMSE). These metrics quantify different dimensions of forecast accuracy, including the average magnitude of errors and the variability of model performance across observations.

Mean Squared Error (MSE) The MSE evaluates the average of the squared differences between predicted and actual values. It is defined in Equation 23:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{23}$$

where:

- n is the number of observations,
- y_i is the actual value at time i,
- $\hat{y}_i$ is the predicted value at time i,
- $(y_i - \hat{y}_i)^2$ is the squared prediction error for observation i.

MSE penalizes larger errors more heavily due to squaring, making it sensitive to outliers. It is useful when large deviations from actual values are particularly undesirable.

Root Mean Squared Error (RMSE) The RMSE is the square root of the MSE and expresses the error in the same units as the original variable. It is calculated using Equation 24:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{24}$$

where:

- All variables are defined as in the MSE,
- The square root ensures the final error is in the original scale of the data.

RMSE emphasizes larger errors and is particularly effective for evaluating models where large forecasting mistakes carry greater risk or cost.

Mean Absolute Error (MAE) The MAE computes the average magnitude of prediction errors without considering their direction. It is defined by Equation 25:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{25}$$

where:

- $|y_i - \hat{y}_i|$ represents the absolute error for observation i,
- Other variables are as previously defined.

MAE provides a straightforward interpretation of average error size and is less sensitive to outliers compared to MSE and RMSE.

Heteroscedastic Mean Squared Error (HMSE) The HMSE evaluates prediction accuracy while accounting for time-varying volatility in the target series. Instead of treating all errors equally, it scales each squared error by the level of the actual value, yielding a relative error measure. HMSE is defined by Equation 26:

$$\text{HMSE} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{y_i}\right)^2 \tag{26}$$

where:

- y_i is the actual realized volatility at time i,

- $\hat{y}_i$ is the predicted realized volatility at time i,
- The squared ratio reflects the relative prediction error at each time point.

This metric is particularly appropriate for financial time series, where the magnitude of volatility can vary substantially. By evaluating errors in proportion to the actual volatility level, HMSE offers a scale-invariant assessment of model performance under heteroscedastic conditions.

5 Results and Error Metrics

5.1 Evaluation of GARCH Model Performance

The predictive performance of the GARCH(1,1) model is visually assessed in Figure 7, which compares the model's one step ahead realized volatility forecasts to the historical realized volatility of the S&P 500 index. The forecasts are generated using a rolling window of 252 trading days, and as such, the plot spans the period from January 4, 2005 which is the first out-of-sample prediction, through January 1, 2024. In the figure, realized volatility is depicted in dark orange, while the GARCH(1,1) forecasts are shown in black.

The visualization reveals that the GARCH(1,1) model effectively captures the general trend and cyclical patterns of realized volatility, including notable spikes corresponding to periods of market stress. However, the model systematically underestimates the magnitude of realized volatility across the full forecast horizon. This underprediction is a well-documented limitation of standard GARCH models, which often struggle to fully accommodate the tail behavior and sharp volatility bursts characteristic of financial markets.

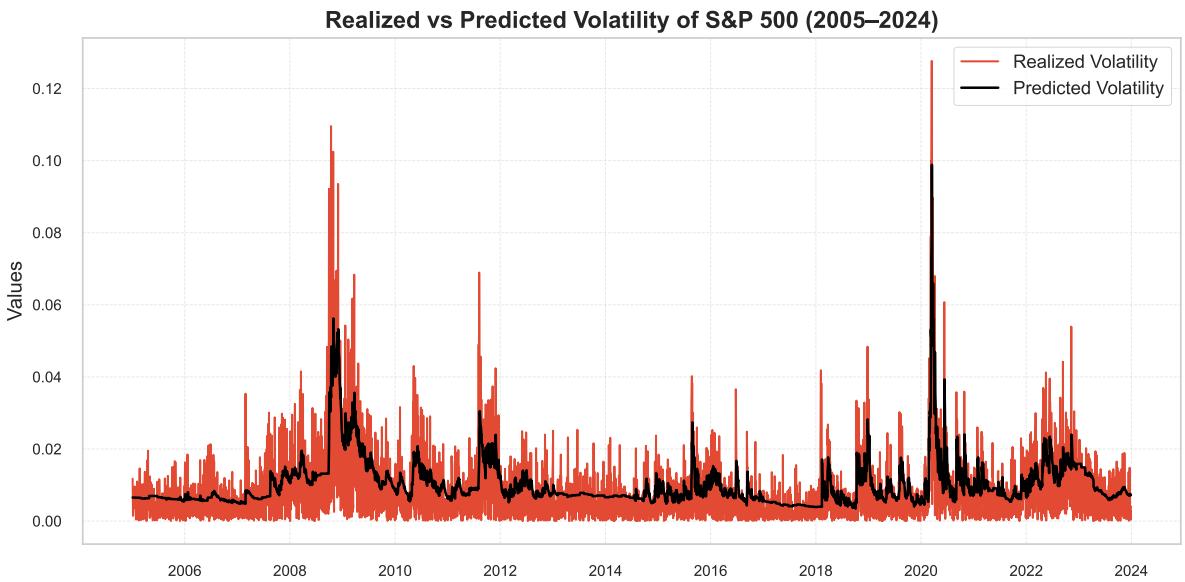

Figure 7: Realized and predicted volatility of the S&P 500 from Jan 2005 to Jan 2024. Source: data from Yahoo Finance

Table 3 presents the performance of the volatility forecasting of GARCH(1,1) model based on the error metrics discussed previously, namely the Mean Squared Error (MSE), Root Mean

Squared Error (RMSE), Mean Absolute Error (MAE), and Heteroscedastic Mean Squared Error (HMSE). These metrics provide a comprehensive assessment of model accuracy by capturing both the magnitude and variability of forecast errors. The error metrics were calculated over the out-of-sample forecast interval from January 9, 2009, to December 29, 2023, aligning with the period during which the LSTM model's out-of-sample forecasts were produced. This consistent evaluation window ensures a fair and robust comparison of performance across all three models, enabling a clearer assessment of their relative effectiveness under varying market conditions.

According to the results, the Mean Squared Error (MSE) was approximately 0.6398, indicating the average squared deviation between the predicted and actual realized volatility. The Root Mean Squared Error (RMSE), which retains the original unit of measurement and emphasizes larger errors due to the square root transformation, was recorded at 0.7999. The Mean Absolute Error (MAE), a more robust metric that is less sensitive to outliers, was 0.5912. These values suggest that the model achieved reasonable predictive accuracy across the forecast horizon. Notably, the HMSE value was approximately 8.3×10^{11}, which reflects the model's performance in the presence of heteroscedasticity, an inherent characteristic of financial time series. The significantly larger scale of HMSE compared to the other metrics arises from the fact that it incorporates time-varying volatility into its error computation, amplifying the penalty for mispredictions during periods of market stress or high variance.

Overall, these results suggest that the model, on average, demonstrates reasonable accuracy in predicting realized volatility under typical market conditions. However, the magnitude of the error metrics indicates that there remains considerable room for improvement, especially in accurately capturing extreme volatility episodes. These findings highlight the potential value of more flexible or data-driven approaches that can better adapt to the nonlinear and time-varying structure of financial volatility.

Metric	**GARCH(1,1)**
MSE	0.6398
RMSE	0.7999
MAE	0.5912
HMSE	8.3×10^{11}

Table 3: Evaluation metrics for GARCH(1,1)-predicted volatility vs actual realized volatility.

5.2 Evaluation of LSTM Model Performance

The performance of the Long Short-Term Memory (LSTM) network is illustrated in Figure 8, which plots the one-step-ahead forecasts against the corresponding realized volatility values. The forecasts were generated using a walk-forward validation approach with a training window of 1,008 observations, a validation split of 33%, and an out-of-sample test window of 504 daily observations, except for the final window, which was limited to 241 observations due to data constraints. The LSTM model's out-of-sample predictions begin on January 9, 2009, and continue until December 29, 2023. As shown in the figure, the historical realized volatility is represented by a dark orange line, while the LSTM's predicted daily realized volatility is depicted in dark purple. Visually, the LSTM model demonstrates a notable improvement over the classical GARCH(1,1) approach, particularly in its ability to capture short-term volatility fluctuations and more accurately reflect the magnitude of volatility spikes. The LSTM model showcased better ability to adjust to changes in realized daily volatility over time, respond-

ing more effectively to short-term movements and local shifts than the classical GARCH(1,1) model. However, it still exhibits a tendency to underpredict during periods of heightened market turbulence.

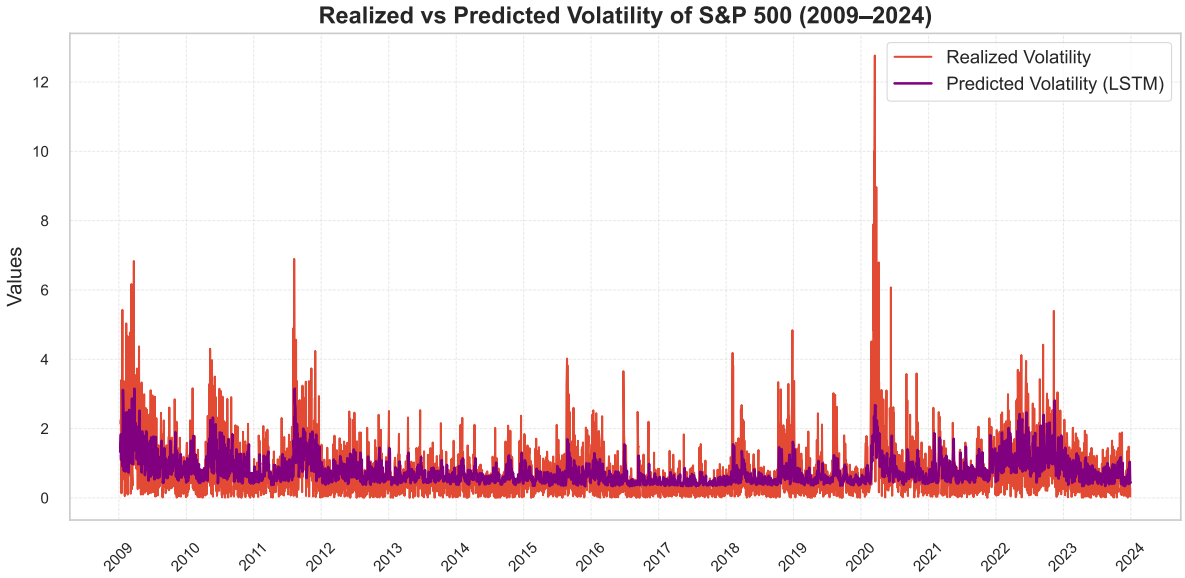

Figure 8: Realized and predicted volatility of the S&P 500 from Jan 2009 to Jan 2024. Source: data from Yahoo Finance

Table 4 reports the performance metrics for the LSTM model's one-step-ahead forecasts of realized volatility over the out-of-sample period from January 9, 2009, to December 29, 2023. The evaluation includes four commonly used error metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Heteroscedastic Mean Squared Error (HMSE). The LSTM model achieved the following results: MSE = 0.6268, RMSE = 0.7917, MAE = 0.5371, and HMSE = 4.0×10^{11}.

These results indicate a clear improvement in predictive accuracy across all metrics when compared to the classical GARCH(1,1) model. Notably, the HMSE, which accounts for the heteroscedastic nature of volatility, shows a substantial reduction of over 50% in relation to the GARCH model, suggesting that the LSTM was significantly more effective at capturing the varying error structure often present in financial time series. While the reductions in MSE, RMSE, and MAE are less dramatic than that of HMSE, they are nonetheless meaningful and point to an overall enhancement in model performance.

Metric	**LSTM**
MSE	0.6268
RMSE	0.7917
MAE	0.5371
HMSE	4.0×10^{11}

Table 4: Evaluation metrics for LSTM-predicted volatility vs actual realized volatility.

5.3 GARCH-LSTM Ensemble

The GARCH-LSTM ensemble model was developed by dynamically combining the one-step-ahead forecasts produced by the LSTM network and the GARCH(1,1) model, using the inverse-error weighting method described earlier. This approach allows the ensemble to adaptively adjust the contribution of each model over time, depending on their relative forecasting performance.

Figure 9 illustrates the ensemble model's one-day-ahead realized volatility forecasts over the out-of-sample period from January 2009 to January 2024. The actual realized volatility is represented by a dark orange line, while the ensemble's predicted volatility is shown in dark blue.

Visual inspection of the figure suggests that the ensemble model seems to outperform the individual models in capturing both sharp volatility spikes and subtle short-term fluctuations. Specifically, the ensemble appears to inherit the strengths of both constituent models: it seems to perform better than the GARCH(1,1) model in detecting rapid changes during low-volatility regimes and better than the LSTM model in its responsiveness to extreme volatility events. This balance enables the ensemble to offer a more accurate and consistent representation of market volatility across different market conditions.

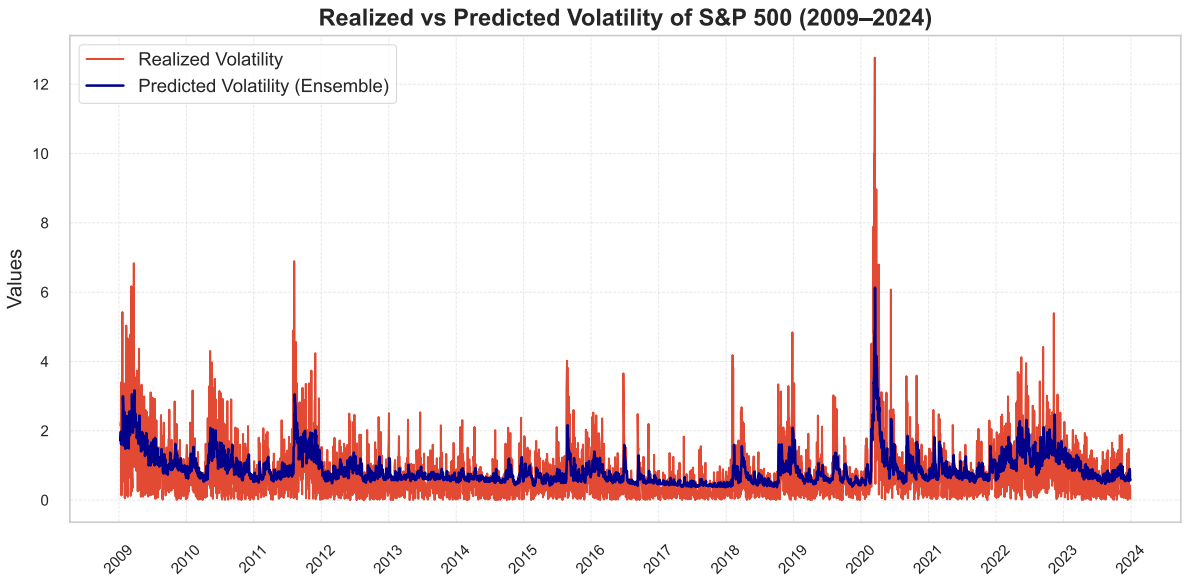

Figure 9: Realized and predicted volatility of the S&P 500 from Jan 2009 to Jan 2024. Source: data from Yahoo Finance

Table 5 presents the dynamically calculated weights for the GARCH and LSTM models across each of the eight rolling windows, as determined by the inverse-error weighting method described in Section 4.3.1. These weights reflect the relative predictive performance of the two models within each window and are used to compute the ensemble forecasts of one-step-ahead realized volatility.

Table 6 presents the performance of the GARCH-LSTM ensemble model in forecasting realized volatility, evaluated using the four key error metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Heteroscedastic Mean

Window	w_{LSTM}	w_{GARCH}
1	0.500000	0.500000
2	0.519643	0.480357
3	0.522546	0.477454
4	0.527902	0.472098
5	0.522115	0.477885
6	0.522755	0.477245
7	0.519813	0.480187
8	0.520990	0.479010

Table 5: Per-window weights assigned to LSTM and GARCH models over the eight out-of-sample test windows.

Squared Error (HMSE). These metrics were computed over the out-of-sample forecast period from January 9, 2009, to December 29, 2023, corresponding to the interval during which the LSTM model's forecasts were also evaluated.

The ensemble model achieved the following results: MSE = 0.5774, RMSE = 0.7598, MAE = 0.5472, and HMSE = 5.8×10^{11}. These values indicate that the ensemble approach outperformed both the standalone GARCH(1,1) and LSTM models in terms of MSE and RMSE, demonstrating improved overall accuracy and lower average squared prediction error.

While the MAE and HMSE values were slightly higher than those obtained using the LSTM model alone, they remained lower than the corresponding metrics for the GARCH(1,1) model. This suggests that although the LSTM model performed best in minimizing average absolute error and accounting for heteroscedasticity, the ensemble achieved a more balanced performance across all metrics.

Metric	GARCH-LSTM
MSE	0.5774
RMSE	0.7598
MAE	0.5472
HMSE	5.8×10^{11}

Table 6: Evaluation metrics for GARCH-LSTM predicted volatility vs actual realized volatility.

5.4 VaR and ES

In this subsection, the results of the Value at Risk and Expected Shortfall predictions are discussed. Figure 10 illustrates the Value at Risk (VaR) estimates produced by the three models, namely GARCH(1,1), LSTM, and their dynamic ensemble at the 95% confidence level. These VaR forecasts are plotted alongside the actual daily return observations over the out-of-sample period from January 2009 to January 2024, which contains 3769 observations, to evaluate each model's ability to capture downside risk. The figure shows that all three models produce VaR estimates that generally follow the dynamics of market returns, adjusting to changing volatility conditions over time. Instances where the actual return falls below the predicted VaR threshold are marked with colored dots, representing violations or VaR breaches, cases where the model underestimated the risk and failed to capture the observed loss. The GARCH(1,1) forecasts

and violations are shown in red, LSTM results are in yellow, ensemble model outcomes are illustrated in green, and actual returns are represented in grey.

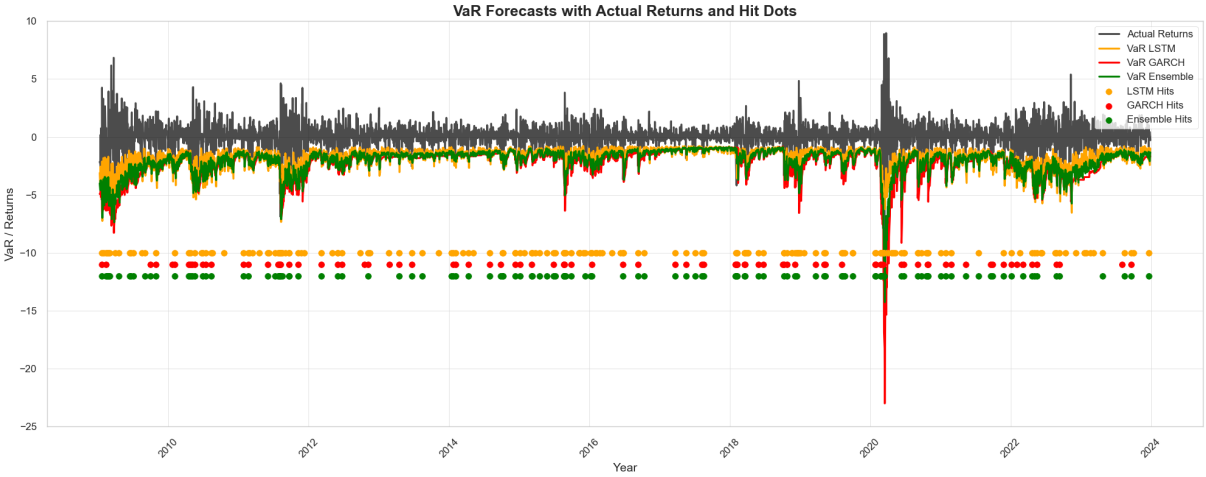

Figure 10: Visualization of Value at Risk Predictions (VaR) for GARCH(1,1), LSTM and GARH-LSTM ensemble against daily returns with violations illustrated as dots.

Table 7 presents a concise summary of the Value at Risk (VaR) and Expected Shortfall (ES) forecasts generated by each model, as well as the corresponding hit rates. The hit rate indicates the proportion of VaR breaches. According to the results, the GARCH(1,1) model consistently produced the most conservative risk estimates, with its average VaR and ES values being the most negative among the three models at -1.6518 and -1.5832, respectively. The ensemble model followed, offering moderately conservative estimates with mean VaR at -1.4502 and ES at -1.8186, while the LSTM model was the least conservative, yielding comparatively less negative average VaR and ES values of -1.2624 and -1.5832, respectively. In terms of hit rates, the GARCH(1,1) model recorded breaches in 5.73% of cases, closely aligning with the nominal 5% level, the LSTM model, by contrast, produced a higher hit rate of 9.58%, indicating frequent underestimation of tail risk and less reliable risk coverage. The ensemble model demonstrated a hit rate of 7.26%, suggesting improved calibration relative to the LSTM, while still producing slightly higher breaches than classical GARCH(1,1).

Model	**Mean VaR**	**Std VaR**	**Mean ES**	**Std ES**	**Hit Rate**
LSTM	-1.2624	0.6304	-1.5832	0.7906	0.0958
GARCH	-1.6518	1.0053	-2.0714	1.2606	0.0573
Ensemble	-1.4502	0.7538	-1.8186	0.9453	0.0726

Table 7: Updated summary statistics for 1% Value-at-Risk (VaR) and Expected Shortfall (ES) forecasts, including mean and standard deviation across the out-of-sample period. Hit rate indicates the empirical frequency of VaR violations.

To assess the accuracy and reliability of the Value-at-Risk (VaR) forecasts, a backtesting procedure was conducted using two widely adopted statistical tests: the Kupiec Proportion of Failures (POF) test and the Christoffersen Conditional Coverage test. The results are reported in Table 8.

The Kupiec test evaluates unconditional coverage by comparing the observed proportion of VaR violations, instances where actual losses exceed the predicted VaR, with the expected proportion, given the model's confidence level. The null hypothesis states that the violation rate is consistent with the model's stated coverage level. As shown in the table, all three models: LSTM, GARCH(1,1), and the ensemble fail the Kupiec test at the 5% significance level, with p-values below 0.05. This indicates that the actual frequency of violations deviates from expectations, suggesting that each model systematically underestimates risk.

The Christoffersen framework extends the analysis by introducing two additional components: the Independence test, which checks whether VaR violations occur randomly over time, meaning that they are independently distributed, and the Conditional Coverage test, which jointly assesses both the correctness of the violation rate and the independence of violations.

According to the results of the Independence test, the null hypothesis of independence in VaR violations is rejected for the LSTM model at the 5% significance level with a p-value of 0.0159, indicating evidence of autocorrelation or clustering in the occurrence of violations. In contrast, there is no reason to reject the null hypothesis for the GARCH(1,1) and ensemble models, which yield p-values of 0.8533 and 0.1580, respectively. These results suggest that, for these two models, the timing of violations is consistent with the independence assumption.

The Conditional Coverage test, which jointly evaluates the accuracy of the violation frequency and the independence of violations over time, reinforces these findings. For both the LSTM and ensemble models, the null hypothesis of correct conditional coverage is rejected at the 5% significance level with p-values < 0.05, indicating potential misspecification in either the expected violation rate, the independence of violations, or both. In contrast, for the GARCH(1,1) model, there is no reason the reject the null hypothesis since p-value = 0.1293, suggesting that its VaR forecasts are more consistent with the assumptions of correct coverage and temporal independence.

Overall, this backtesting analysis reveals that while none of the models achieves statistically valid unconditional coverage, the GARCH(1,1) model demonstrates the most consistent performance in terms of independence and conditional adequacy of VaR violations. The ensemble as well as LSTM models, despite showing improved accuracy in conventional error metrics, do not exhibit superior performance in VaR backtesting. This suggests that enhancements in point forecast accuracy do not necessarily translate into more reliable tail risk estimation. While the ensemble and LSTM models offer potential advantages in other predictive contexts, their VaR forecasts require further refinement to meet the requirements of financial risk management.

Model	**Kupiec LR (p-value)**	**Indep. LR (p-value)**	**CondCov LR (p-value)**
LSTM	132.685 (0.000)	5.812 (0.0159)	138.497 (0.000)
GARCH	4.057 (0.0440)	0.034 (0.8533)	4.091 (0.1293)
Ensemble	36.074 (0.000)	1.993 (0.1580)	38.067 (0.000)

Table 8: Kupiec (POF), Independence, and Conditional Coverage (Christoffersen) test statistics and p-values for evaluating the adequacy of Value-at-Risk (VaR) forecasts. The Kupiec test assesses the accuracy of violation frequency; the Independence test checks whether violations occur randomly over time; and the Conditional Coverage test jointly evaluates both criteria.

To evaluate the accuracy of Expected Shortfall (ES) forecasts, the McNeil and Frey [2000]

backtesting procedure was applied. This test assesses whether the conditional mean of losses exceeding the VaR threshold aligns with the model's ES forecasts. The null hypothesis states that the ES is correctly specified, meaning that the average loss beyond the VaR level corresponds to the predicted ES. The test uses a t-statistic that measures the difference between the model's predicted Expected Shortfall (ES) and the actual average losses observed beyond the Value at Risk (VaR) threshold. These losses are known as exceedances. If the t-statistic is significantly different from zero, it indicates that the model's ES forecasts are not accurately capturing the true magnitude of extreme losses, leading to a rejection of the hypothesis that the ES is correctly specified.

As shown in Table 9, all three models, namely LSTM, GARCH(1,1), and the ensemble, yielded highly significant t-statistics: 35.439, 28.149, and 29.823, respectively, with p-values less than 0.05. These results lead to a rejection of the null hypothesis at the significance level of 5% for all models, indicating that none of the ES forecasts are statistically consistent with the realized losses in the tail of the distribution. The number of exceedances associated with each model is also reported for context, with the LSTM model exhibiting the highest number of 361, followed by the ensemble, which showed 274, and GARCH(1,1) illustrated 216, which is well higher than the expected number of 189.

These findings suggest that, despite differences in exceedance frequency, all models exhibited systematic errors in estimating tail risk severity, as captured by the ES measure. Thus, while VaR violations may be more accurately timed in some models (e.g., GARCH), the models collectively fall short in quantifying the magnitude of losses beyond the VaR threshold.

Model	***t*-statistic**	**p-value**	**Exceedances**
LSTM	35.439	0.000	361
GARCH	28.149	0.000	216
Ensemble	29.823	0.000	274

Table 9: McNeil and Frey (2000) test statistics for evaluating the accuracy of Expected Shortfall (ES) forecasts. The test is based on exceedances (observations beyond the VaR threshold). A significant t-statistic implies a rejection of the null hypothesis that the ES is correctly specified.

6 Conclusion

This thesis evaluated the out-of-sample forecasting performance of three volatility forecasting models, namely GARCH(1,1), Long Short-Term Memory (LSTM), and a dynamically weighted GARCH-LSTM ensemble using daily S&P 500 return data spanning the period from 2004 to 2024. Model performance was assessed not only through conventional accuracy metrics like Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Heteroscedastic Mean Squared Error (HMSE) but also in terms of their effectiveness in estimating market risk via Value-at-Risk (VaR) and Expected Shortfall (ES). Risk model reliability was further validated through formal backtesting using the Kupiec Proportion of Failures (POF) test and the Christoffersen Conditional Coverage test, as well as the McNeil and Frey test.

The results lead to several key conclusions:

- **RQ1:** although the LSTM model achieved lower error values than classical GARCH(1,1) across all point forecast metrics, it performed poorly in risk evaluation tests. Specifically, it had the highest number of VaR violations and failed both the Kupiec and Christoffersen independence and conditional coverage tests. This indicates that better point forecast accuracy does not necessarily translate into improved risk prediction or tail loss modeling.
- **RQ2:** the dynamically weighted ensemble model outperformed both the GARCH(1,1) and LSTM models in terms of error metrics, showing improved accuracy in capturing daily return volatility. However, its performance in risk diagnostics was mixed: while it had fewer violations than LSTM and passed the independence test, it failed both the Kupiec and Christoffersen conditional coverage tests, as well as the Expected Shortfall backtest.
- **RQ3:** the ensemble model did not produce the closest to expected α rate of VaR violations - the GARCH(1,1) model did. Among the three models, GARCH(1,1) recorded the closest number of VaR breaches to the expected level and was also the only model to pass the Christoffersen conditional coverage test. Although it, like the others, failed the Kupiec and Expected Shortfall (ES) backtests, it came closest to satisfying all the backtesting criteria. These results indicate that, despite its simplicity and relatively higher forecast errors, the GARCH(1,1) model remains the most statistically reliable option for estimating tail risk in this study.

Overall, none of the evaluated models, namely GARCH(1,1), LSTM, or the dynamically weighted ensemble, were able to fully meet the statistical backtesting requirements for both Value-at-Risk (VaR) and Expected Shortfall (ES). Although the LSTM and ensemble models provided better accuracy in forecasting realized volatility based on standard error metrics, this improvement did not translate into better estimation of extreme losses. A key reason for this outcome is that the models were trained to minimize average forecast errors, such as MSE or MAE, which do not specifically focus on rare but impactful tail events. In contrast, the GARCH(1,1) model, while less precise in day-to-day volatility prediction, showed more reliable performance in risk estimation, especially in terms of VaR exceedance stability. This likely reflects its simpler and more stable structure, which aligns better with risk-focused metrics despite lower point forecast accuracy.

These findings highlight an important limitation in the current model setup: improving general accuracy does not necessarily improve tail risk prediction. To address this, future research may consider modifying the model training objectives to put more weight on large forecast errors, for example, by optimizing for quantile-based loss functions that emphasize tail errors to focus training on extreme forecast deviations. Alternatively, models could be trained with metrics that directly reflect risk performance to better capture and manage downside risk. Moreover, the ensemble framework itself may benefit from refining the weights assignment approach to emphasize precise estimation of extreme losses rather than merely minimizing average errors. Such improvements would help create models that are not only more adaptive but also more aligned with the practical needs of financial risk management.

Several limitations of this study should be acknowledged. Firstly, the calculation of Value-at-Risk (VaR) in this thesis is based on the standard parametric approach, where VaR at a given confidence level α is computed as $z_\alpha \cdot \sigma_t$, with z_α denoting the corresponding quantile of the standard normal distribution and σ_t representing the model-specific forecasted volatility. This formulation implicitly assumes that the standardized residuals from all models, including GARCH, LSTM, and the ensemble, follow a standard normal distribution. While this assumption sim-

plifies estimation and interpretation, it does not adequately capture the empirical features of financial return distributions, particularly fat tails and potential skewness. Consequently, the resulting VaR estimates may underestimate extreme losses, especially during periods of high market volatility or financial stress. Another important point is that the LSTM model in this study was trained using realized volatility, which was approximated by squared daily logarithmic returns. This approach assumes that realized volatility is a good estimate of the actual, unobserved volatility. While this method is widely used because it's straightforward and based on available data, it's not perfect. Squared returns can be influenced by random noise and might not always reflect true volatility, especially during times of sudden price jumps or irregular market behavior. This means that the model might sometimes be learning from a noisy or imperfect target, which could affect the accuracy of its predictions. Secondly, although the LSTM model was tuned using a randomized search strategy, it is possible that the model's performance could be further improved by using more advanced hyperparameter optimization methods. Techniques such as Bayesian optimization or evolutionary algorithms might explore the hyperparameter space more efficiently and lead to more accurate forecasting outcomes. Third, the dynamic weighting scheme used in the GARCH-LSTM ensemble is based on recent values of the Mean Absolute Error (MAE) as a proxy for model performance. While this method works reasonably well under relatively stable conditions, it may become less reliable during times of structural breaks or sudden regime changes in market volatility. In such cases, recent forecasting performance may not be a good indicator of future accuracy.

For future work, several improvements could be considered. One possible direction would be to relax the assumption of normally distributed GARCH innovations by estimating the model using alternative error distributions, such as the Student's t-distribution, generalized error distribution (GED), or skewed distributions. These alternatives are more suitable for capturing the heavy tails and asymmetries typically observed in financial return series, and may lead to more accurate estimation of Value-at-Risk (VaR) and Expected Shortfall (ES), especially during periods of market stress. Furthermore, the predictive performance and adaptability of the ensemble model could potentially be improved by integrating more advanced architectures into the framework, such as attention-based neural networks or transformer models. These architectures may offer enhanced capability in modeling complex temporal dependencies and regime shifts in financial time series. Finally, the risk evaluation framework could be extended beyond traditional measures such as VaR and ES. Incorporating additional tools such as stress testing, scenario analysis, or systemic risk indicators would provide a more comprehensive understanding of market risk exposure, particularly under extreme or crisis conditions.

To conclude, this thesis showed that while machine learning models such as Long Short-Term Memory (LSTM) networks provided better average forecast accuracy, they did not yield more reliable risk estimates in terms of Value-at-Risk (VaR) and Expected Shortfall (ES). The GARCH(1,1) model, despite being less accurate in point forecasting, demonstrated more consistent performance in regulatory risk backtests. The dynamically weighted ensemble offered a useful compromise by improving volatility forecast precision, though its risk estimation capabilities remain limited and did not outperform GARCH in statistical validation. These results highlight the trade-off between forecast accuracy and risk reliability in financial modeling. The findings support further exploration of hybrid approaches that combine the strengths of econometric and machine learning models while being more explicitly tailored to tail risk evaluation.

References

T. G. Andersen, T. Bollerslev, F. X. Diebold, and P. Labys. Modeling and forecasting realized volatility. *Econometrica*, 71(2):579–625, 2003.

D. Ardia and L. F. Hoogerheide. Garch models for daily stock returns: Impact of estimation frequency on value-at-risk and expected shortfall forecasts. *Economics Letters*, 123(2):187–190, 2014.

Y. Bengio, P. Simard, and P. Frasconi. Learning long-term dependencies with gradient descent is difficult. *IEEE Transactions on Neural Networks*, 5(2):157–166, 1994.

J. Bergstra and Y. Bengio. Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 13(Feb):281–305, 2012.

F. Black. Studies of stock price volatility changes. *Proceedings of the 1976 Meetings of the American Statistical Association, Business and Economic Statistics Section*, pages 177–181, 1976.

T. Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

T. Bollerslev, R. Y. Chou, and K. F. Kroner. Arch modeling in finance: A review of the theory and empirical evidence. *Journal of Econometrics*, 52(1–2):5–59, 1992.

A. Borovykh, S. Bohte, and C. W. Oosterlee. Conditional time series forecasting with convolutional neural networks. *arXiv preprint arXiv:1703.04691*, 2017.

P. F. Christoffersen. Evaluating interval forecasts. *International Economic Review*, 39(4):841–862, 1998.

R. F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.

J. C. Gamboa. Deep learning for time-series analysis. *arXiv preprint arXiv:1701.01887*, 2017.

L. R. Glosten, R. Jagannathan, and D. E. Runkle. On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5): 1779–1801, 1993.

S. Hochreiter and J. Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.

S. Hosseini and M. Kiani. A survey on the application of lstm networks in financial forecasting. *Journal of Computational Finance*, 26(1):45–68, 2022. doi: 10.21314/JCF.2022.448.

P. Jorion. *Value at Risk: The New Benchmark for Managing Financial Risk*. McGraw-Hill, New York, 3rd edition, 2007.

I. Kaastra and M. Boyd. Designing a neural network for forecasting financial and economic time series. *Neurocomputing*, 10(3):215–236, 1996.

H. Kim and C. Won. Forecasting the volatility of stock price index: A hybrid model integrating lstm with multiple garch-type models. *Expert Systems with Applications*, 103:25–37, 2018.

W. Kristjanpoller and M. C. Minutolo. Forecasting volatility of oil price using an artificial neural network–garch model. *Expert Systems with Applications*, 65:233–241, 2016.

P. H. Kupiec. Techniques for verifying the accuracy of risk measurement models. *The Journal of Derivatives*, 3(2):73–84, 1995.

H. Liu. Forecasting stock volatility with garch, svr and lstm models: An empirical study. *Procedia Computer Science*, 147:400–406, 2019.

A. J. McNeil and R. Frey. Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach. *Journal of Empirical Finance*, 7(3–4):271–300, 2000. doi: 10.1016/S0927-5398(00)00012-8.

A. J. McNeil, R. Frey, and P. Embrechts. *Quantitative Risk Management: Concepts, Techniques, and Tools*. Princeton University Press, revised edition, 2015. ISBN 9780691166278.

D. B. Nelson. Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2):347–370, 1991.

Basel Committee on Banking Supervision. Minimum capital requirements for market risk. Technical report, Bank for International Settlements, January 2016. URL `https://www.bis.org/publ/bcbs265.pdf`. Available at: `https://www.bis.org/publ/bcbs265.pdf`.

OpenAI. ChatGPT Plus (May 2025 version). `https://chat.openai.com`, 2025. Used for language rephrasing and grammar refinement in thesis preparation.

O. B. Sezer, M. U. Gudelek, and A. M. Ozbayoglu. Financial time series forecasting with deep learning: A systematic literature review: 2005–2019. *Applied Soft Computing*, 90:106181, 2020.

S. J. Taylor. *Modelling Financial Time Series*. World Scientific, 2nd edition, 2008.

R. S. Tsay. *Analysis of Financial Time Series*. Wiley, 2nd edition, 2005.

L. Yu and X. Li. A deep learning-based lstm approach for forecasting volatility in financial markets. *Neurocomputing*, 356:240–250, 2018.

J. Zhang, H. Liang, Q. Zhu, and Y. Li. Combining deep learning and traditional models for financial volatility forecasting: Empirical evidence from lstm and tcn architectures. *Journal of Forecasting*, 41(1):139–154, 2022.